

1. **Application of Human Activity/Action Recognition: A Review** (*Multimedia Tools and Applications*, Springer, 2025)

Focus: Comprehensive breakdown of computer vision frameworks, video-based pattern recognition, and behavior classification algorithms for tracking user actions.

Link -

https://www.researchgate.net/publication/387824910_Application_of_human_activityaction_recognition_a_reviewhttps://www.researchgate.net/publication/387824910_Application_of_human_activityaction_recognition_a_review

Overview

Human Activity Recognition (HAR) is a vital domain in computer science and artificial intelligence focused on detecting, classifying, and predicting human behaviors using data collected from sensors or vision-based systems. With computational power advancements, deep learning algorithms have become standard for addressing HAR problems. This research presents a comprehensive literature review of 80 sources (categorized into 42 core references) that analyze deep learning and machine learning models in the HAR domain, specifically emphasizing the distinction between two core concepts: **activity** (continuous, related affairs over time) and **action** (short-term, specific movements and behaviors).

Project Objective

The primary objectives of this review study are to:

- Explore and categorize various sensor types (wearable, environmental, and object-based) and vision-based technologies utilized in HAR.
- Investigate the application of traditional machine learning and advanced deep learning architectures (such as CNNs, RNNs, LSTMs, and DNNs) used across different frameworks.
- Differentiate between human activity recognition and human action recognition by analyzing their respective data acquisition methods, focus areas, and datasets.
- Identify lifestyle applications, current challenges, and potential future research directions to improve human well-being, healthcare, and security systems.

Research Gap

Through an analysis of previous review articles, the paper notes that existing literature fails to draw a clear line between the branches of activity recognition and action recognition. Most prior surveys lump broad, continuous lifestyle activities and short-term, specific physical movements

together under a single umbrella. Furthermore, previous studies often lack a granular breakdown mapping specific sensor types, distinct dataset categories, and unique algorithmic architectures side-by-side for both modes.

Research Questions

While the paper acts as a narrative and systematic review rather than listing explicit question statements, the foundational questions guiding the research can be synthesized as:

- How can human behavior monitoring be systematically distinguished between continuous "activities" and short-term "actions"?
- What are the strengths, limitations, and classification accuracies of various machine learning versus deep learning algorithms when applied to sensor- and vision-based HAR data?
- Which datasets, sensor configurations, and architectural pipelines are best suited for specific real-world applications such as healthcare, elderly care, and smart surveillance?

Key Contributions

The primary contributions of this research paper include:

- **Clear Conceptual Separation:** Distinctly separating and defining the scope of articles into two categories: **human activity** (daily routines, professional tasks) and **human action** (short-term body movements, social actions).
- **Differentiated Categorization:** Systematically specifying sensor types and activity/action types separately for both modes.
- **Dataset Breakdown:** Providing an organized breakdown of prominent public datasets associated with each state (e.g., UCI-HAR, WISDM, PAMAP2 for activities; ActivityNet, MultiSports, Ego4D for actions).
- **Architecture Mapping:** Identifying and classifying various types of machine learning and deep learning architectures deployed across both activity and action detection frameworks.
- **Lifestyle Application Identification:** Highlighting real-world applications of human activity recognition in improving lifestyle quality, healthcare monitoring, and security.

2..Deep Learning-Based Human Activity Recognition Using Binary Sensor and Vision Fusion (*Applied Sciences*, MDPI, 2026)

Focus: Explores real-time behavior classification systems to distinguish focused activities from idle states or distractions.

Link- <https://www.mdpi.com/2079-9292/15/2/428>

Overview

Human Activity Recognition (HAR) is essential for smart homes and Ambient Assisted Living (AAL) systems, but ambient sensor-based HAR (AHAR) frequently suffers from noisy features and data scarcity. This paper proposes a novel framework that integrates a Deep Learning Ensemble-Based Stacking Neural Network (SNN) with an Ambient Sensor Generative Adversarial Network (ASGAN) to generate synthetic training data. Evaluated across six benchmark datasets from the CASAS smart home project, the proposed stacking framework achieves superior accuracy and robustness compared to traditional state-of-the-art baselines.

Project Objective

The primary objective of this research is to develop a unified, privacy-preserving AHAR framework that simultaneously addresses two critical challenges in real-world smart home deployments:

- Overcoming feature extraction limitations and noisy data by leveraging ensemble-based stacking of diverse spatial and temporal deep learning models.
- Mitigating data scarcity and class imbalance through adversarial data augmentation using a specialized Generative Adversarial Network (ASGAN).

Research Gap

The paper identifies that existing AHAR systems face several core limitations:

- Traditional machine learning models rely heavily on manual feature engineering, while individual deep learning models (such as isolated CNNs or LSTMs) fail to jointly capture the complementary spatial and temporal characteristics required for robust recognition.
- Many state-of-the-art models assume the availability of sufficiently large, well-balanced datasets and oracle-segmented data, which are rarely available in continuous, real-world smart home deployments where labeled data is scarce.

Research Questions

Although presented implicitly through the methodology, the core questions driving the research include:

- How can heterogeneous deep learning base models be effectively integrated via stacking to capture both spatial abstraction and long-range temporal dependencies from sparse, binary ambient sensor streams?
- Can generative adversarial networks be designed to successfully capture the temporal continuity, sparsity, and event-driven nature of ambient sensor data to alleviate data scarcity?

- To what extent do architectural, data, and sliding window size diversities enhance the generalization and accuracy of stacking-based AHAR models across diverse smart home environments?

Key Contributions

The main contributions presented in this research are:

- **Deep Stacking Neural Network (SNN):** Proposing an ensemble-based architecture that systematically integrates diversified CNN and LSTM base models to jointly capture spatial and temporal characteristics of ambient sensor data.
- **Ambient Sensor Generative Adversarial Network (ASGAN):** Introducing an LSTM-based GAN designed to synthesize realistic ambient sensor sequences to address data scarcity and enhance model performance through adversarial data augmentation.
- **Rigorous Empirical Evaluation:** Conducting extensive experiments on six CASAS smart-home datasets, demonstrating statistically significant accuracy improvements (averaging a 13.28 percentage point gain) over representative state-of-the-art baselines using non-parametric statistical tests.
- **Ensemble Diversity Analysis:** Analyzing the impact of architectural diversity, training data diversity, and sliding window size diversity within stacking frameworks to provide practical insights for ensemble design.

3A Review of Recent Techniques for Human Activity Recognition (*Algorithms*, MDPI, 2024)

Focus: Analyzes how modern multimodal networks and computer vision models track human actions in indoor workspace environments.

Link- <https://www.mdpi.com/1999-4893/17/10/434>

Overview

- Human Activity Recognition (HAR) is a rapidly evolving domain focused on automatically detecting and recognizing human behavior using sensory information from the body and environment.
- This survey paper provides a comprehensive overview of the state-of-the-art in HAR, with a specific focus on recent technological advancements including multimodality, Deep Reinforcement Learning (DRL), and large language models (LLMs).

Project Objective

- To explore the diverse range of human activities and sensor technologies used for data collection.
- To review novel algorithms with an emphasis on multimodality, Deep Reinforcement Learning, and LLMs.
- To provide an overview of multimodal datasets incorporating physiological data.
- To examine the applications of HAR in healthcare, alongside current challenges and future research directions.

Research Gap

- Existing survey papers are often focused on specific, isolated areas (such as smart homes or basic deep learning models) and fail to capture the most recent advancements in multimodality, DRL, and LLMs together.
- Prior reviews lack a discussion on the importance of physiological datasets and the integration of health vital data with traditional sensor data for HAR.

Research Questions

While the paper uses a systematic literature review methodology rather than explicitly listing numbered questions, the core guiding inquiries of the review include:

- How do recent methodologies like multimodality, DRL, and LLMs enhance the accuracy, data annotation efficiency, and personalization of HAR systems?
- What is the role of multimodal datasets containing physiological and inertial data in advancing healthcare-based HAR applications?

Key Contributions

- **Activity and Sensor Analysis:** Provided a structured analysis of human activities and the sensors used for recognition tasks.
- **Literature Review of Vision and Non-Vision HAR:** Reviewed recent vision-based (e.g., skeleton data) and non-vision-based (wearable and ambient sensors) publications.
- **Focus on Modern Algorithms:** Reviewed recent algorithms with a dedicated emphasis on multimodal techniques, DRL, and LLMs.
- **Multimodal Dataset Review:** Provided a detailed overview of multimodal datasets that feature physiological data.

- **Healthcare Applications:** Highlighted and discussed specific real-world applications of HAR in the healthcare sector.
- **Challenges and Future Outlook:** Analyzed existing system challenges and outlined future research directions for the field.

4 Title - A Review of Human Activity Recognition Methods

Focus- Recognizing human activities from video sequences

Link- <https://www.frontiersin.org/journals/robotics-and-ai/articles/10.3389/frobt.2015.00028/full>

Overview

- Human activity recognition from video sequences or still images is a challenging task due to factors such as background clutter, partial occlusions, scale variations, viewpoint changes, lighting, and appearance changes.
- This review paper provides a comprehensive overview of recent and state-of-the-art research advances in human activity classification, categorizing methodologies into unimodal and multimodal approaches, reviewing available datasets, and outlining future research directions.

Project Objective

- To propose a structured hierarchical categorization of human activity recognition methodologies based on the nature of sensor data and underlying behavioral models.
- To analyze and compare the advantages and limitations of various unimodal (space-time, stochastic, rule-based, and shape-based) and multimodal (affective, behavioral, and social networking) approaches.
- To survey existing publicly available human activity classification datasets and detail the requirements for an ideal recognition dataset.
- To identify open research issues and future challenges in unconstrained human behavior understanding.

Research Gap

- Many existing survey papers summarize human activity recognition methods without providing a concise, informative comparison of the strengths and weaknesses of each category.

- Prior research lacks a unified framework that systematically addresses how to bridge the semantic gap between low-level visual features and high-level behavioral concepts (such as emotions and social roles) across different data modalities.

Research Questions

While the paper uses a narrative review approach rather than a formal list of questions, the foundational inquiries guiding the work include:

- How can human activities be systematically categorized according to their complexity (from gestures and atomic actions to group actions and behaviors)?
- What are the performance trade-offs, advantages, and limitations of unimodal versus multimodal human activity recognition methods?
- What characteristics and specifications define an ideal human activity recognition dataset for real-world scenarios?

Key Contributions

- **Hierarchical Categorization:** Proposed a novel hierarchical taxonomy dividing human activity recognition methods into unimodal (space-time, stochastic, rule-based, shape-based) and multimodal (affective, behavioral, social networking) categories.
- **Methodological Comparison:** Provided a detailed analysis outlining the pros and cons of each family of recognition methods.
- **Dataset Classification:** Summarized and categorized prominent public activity recognition datasets across seven distinct groups based on category, year, classes, actors, video clips, and resolution.
- **Open Challenges and Future Directions:** Identified open research problems including pose modeling, handling occlusions, data annotation, and real-time processing in unconstrained environments.

5. Title - Human Activity Recognition: Review, Taxonomy and Open Challenges

Link - <https://www.mdpi.com/1424-8220/22/17/6463>

Overview

Human Activity Recognition (HAR) has become a crucial capability across various domains—including healthcare, smart surveillance, security, and human-computer interaction—driven by the proliferation of Internet of Things (IoT) devices, CCTV systems, and wearable/mobile sensors. Although several surveys have been published previously, the rapidly expanding literature necessitated an updated synthesis. This review surveys 95 selected research articles published between January 2018 and May 2022 to examine contemporary application areas, data sources, algorithmic techniques, and open research challenges in the HAR domain.

Project Objective

The primary objectives of this review study are to:

- Provide a comprehensive state-of-the-art overview of recent HAR literature published since 2018.
- Construct a detailed taxonomy categorizing HAR literature based on application areas, data sources, algorithmic approaches, and open challenges.
- Identify the primary data sources (vision-based vs. sensor-based) and prominent machine learning and deep learning techniques utilized in recent research.
- Highlight current limitations and outline open research challenges to guide future developments in the field.

Research Gap

The paper identifies that while numerous systematic reviews and surveys exist, the majority of them focus narrowly on specific sub-tasks (such as crowd surveillance, fall detection, or healthcare) or target particular data sources (such as wearable sensors alone). There was a lack of an updated, broad review capturing the overall research landscape of HAR irrespective of individual tasks or isolated data sources, particularly considering recent literature.

Research Questions

While the paper uses a systematic literature review methodology rather than listing explicit question statements, the foundational questions steering the research can be synthesized as:

- How can the recent literature on human activity recognition be systematically classified in terms of application areas, data sources, and computational techniques?
- What are the dominant data sources (vision vs. sensor) and algorithmic models (such as CNNs, LSTMs, and SVMs) currently driving HAR advancements?
- What are the primary open challenges and limitations hindering the deployment of robust real-world HAR systems?

Key Contributions

The primary contributions presented in this research review include:

- **Comprehensive Literature Taxonomy:** Generating a structured taxonomy that classifies 95 recent HAR articles across application areas, data sources, algorithmic techniques, and open research challenges.
- **Application Area Distribution:** Revealing that existing research primarily concentrates on daily living activities (42%) and individual/group user activities (34%), with real-time applications such as healthcare, surveillance, and suspicious activity tracking receiving comparatively limited focus (24%).
- **Data Source and Technique Identification:** Identifying that CCTV videos (52%) and mobile sensors (26%) are the most frequently used data sources, while Convolutional Neural Networks (CNNs, 25%), Long Short-Term Memory networks (LSTMs, 13%), and Support Vector Machines (SVMs, 12%) emerge as the most prominent techniques.
- **Open Challenges and Limitations Analysis:** Systematically categorizing and discussing open research hurdles into five key areas: data collection, data preprocessing, hardware and technique constraints, complex activity detection, and activity misalignment.

6. MindEye-OmniAssist: A Gaze-Driven LLM-Enhanced Assistive Robot System for Implicit Intention Recognition and Task Execution

Link - <https://arxiv.org/abs/2503.13250>

Overview

Gaze-based control is a promising approach for human-robot interaction in assistive systems, particularly for individuals with impaired limb functions. However, existing gaze-based systems typically offer limited support constrained to basic grasping actions or predefined scenarios due to restricted intent recognition capabilities. This paper proposes **MindEye-OmniAssist**, an open implicit intention recognition framework and assistive robot system powered by Large Language Models (LLMs) and Vision Foundation Models (VFMs) that processes gaze input to recognize user intents beyond predefined scenarios and executes comprehensive task action sequences.

Project Objective

- To develop an open implicit intention recognition framework using foundation models (LLMs and VFMs) capable of processing natural gaze inputs and addressing undefined user intentions without being restricted to predefined scenarios.

- To implement a complete gaze-driven, LLM-enhanced assistive robot system (MindEye-OmniAssist) that translates recognized user intent into action sequences to help users complete complex, diverse daily tasks.
- To incorporate a user intention confirmation module using intuitive gaze commands to verify inferred tasks before robotic execution.

Research Gap

- Current gaze-based assistive systems are mostly limited to basic grasping functions or rely on finite state machines and preset databases, making them incapable of handling undefined objects or non-predefined tasks.
- Existing implicit intention recognition methods rely heavily on artificially constructed, domain-specific prior knowledge matrices, which restrict generalization and make them difficult to adapt to broader, real-life scenarios.

Research Questions

While the paper adopts a system-building and evaluation methodology rather than formally listing research questions, the guiding inquiries can be synthesized as:

- How can Vision Foundation Models and Large Language Models be integrated with natural gaze tracking to accurately infer open, unconstrained user intentions?
- Can a gaze-driven assistive framework successfully move beyond basic grasping to plan and execute diverse, multi-step assistance tasks (such as pouring water, fetching objects, or toggling switches)?

Key Contributions

- **Open Implicit Intention Recognition Framework:** Developed a novel framework combining an open-vocabulary object detector (YOLO-World), a high-accuracy transformer/convolution-based intention recognition network, and an LLM (DeepSeek-R1) to process gaze inputs and reason about complete user intents.
- **MindEye-OmniAssist System Implementation:** Created an integrated assistive robotic setup featuring head-mounted eye tracking, multi-arm collaborative robots, and multimodal feedback loops that verify intents via voice and gaze-based confirmation interfaces.
- **Rigorous Real-World Validation:** Demonstrated through desktop experiments across 55 trials that the system achieves high intention recognition accuracy and an overall task completion success rate of 41/55 on diverse, undefined tasks.

7. Supporting Productivity Skill Development in College Students through Social Robot Coaching: A Proof-of-Concept

Overview

College students frequently encounter academic stressors and personal transitions that negatively impact their productivity and mental well-being. While self-help books provide generalized guidance and productivity apps offer digital structure, they typically lack educational depth, interactivity, or contextual personalization. Traditional human-led productivity coaching offers personalized support but is resource-intensive and difficult to scale. This study introduces a proof-of-concept for a Socially Assistive Robot (SAR) acting as an educational productivity coach to help college students build soft skills like time management and task prioritization.

Project Objective

- To design and implement a socially assistive robotic coaching system utilizing Large Language Models (LLMs) via prompt engineering to deliver personalized education on time management and task prioritization.
- To provide an interactive web-based dashboard that tracks student metrics (overall progress, mood, confidence per lesson, time spent, and top engagement lessons) and delivers personal productivity insights to foster self-awareness.
- To evaluate the usability, user experience, engagement, and effectiveness of the SAR-based productivity coaching system with undergraduate students.

Research Gap

- Although Socially Assistive Robots (SARs) have been developed for college students and applied in broader coaching contexts, their role in delivering *educational coaching specifically aimed at productivity-related soft skills development* remains underexplored.
- Prior to this work, LLMs had not been formally evaluated in the context of productivity coaching or integrated into SARs for this specific purpose.

Research Questions

While the paper follows a proof-of-concept and evaluation methodology without explicitly listing formal research questions, the guiding goals of the work can be synthesized as:

- Can a socially assistive robot equipped with LLMs effectively serve as an educational productivity coach to address college students' organizational challenges?
- How do students perceive and engage with an SAR-based coaching system that combines verbal/nonverbal robot interactions with an insightful, data-driven tracking dashboard?

Key Contributions

- **LLM-Equipped SAR Productivity Coach:** Developed a Flask-based web and robotic application utilizing GPT-4 and QRobot to deliver six targeted lessons on time

management and task prioritization (such as the Eisenhower matrix, time blocking, and task breakdown).

- **Interactive Monitoring Dashboard:** Integrated a dashboard that tracks metrics like overall progress, sentiment-analyzed mood, time spent per lesson, strategy confidence, and engagement levels while generating tailored personal productivity insights.
- **Positive Empirical Evaluation:** Evaluated the system with 15 undergraduate college students, achieving a System Usability Scale (SUS) score of 79.2 (corresponding to an "A-" grade) alongside high ratings for overall experience (4.3/5) and engagement (4.2/5)

8 Intelligent Technology and Enhanced Well-Being: Can Artificial Intelligence Mitigate Digital Overload?

Overview:

This research studies the relationship between artificial intelligence, employee efficiency, digital overload and well-being. It is particularly relevant because excessive digital technology use can increase workload and create negative experiences even when technology improves efficiency.

Project Objective:

The objective is to understand whether intelligent technologies can develop work efficiency while also helping users manage the negative effects associated with digital overload. It considers the balance between technological benefits and employee well-being.

Research Gap:

AI research frequently emphasizes productivity and efficiency, while the possible consequences or results of excessive technology use receive comparatively decreased attention. More

research is required to understand how AI can support productivity without creating additional digital pressure.

Research Questions:

Can AI improve employee efficiency while supporting well-being? How does digital or techno-overload influence the relationship between AI use and employee well-being?

Key Contribution:

The study provides evidence that AI adoption should be considered together with its effects on users rather than evaluated only through productivity. This supports FocusBot's idea of monitoring work intensity and encouraging appropriate breaks instead of continuously pushing users to work.

9.Interactions with Generative AI: Wearables to Measure Developer Experience and Productivity Objectively

Overview:

This research presents how wearable technology can be used to understand human experiences while interacting with Generative AI. It particularly considers productivity, cognitive load and physiological replies and responses during AI-supported work.

Project Objective:

The objective is to find whether physiological information collected through wearable devices can provide an objective measurement of user experience and productivity. This can complement traditional questionnaires and self-reported productivity measurements.

Research Gap:

Most productivity studies depend heavily on subjective surveys or simple measurements of completed work. These methods may fail to capture cognitive workload, stress and other changes occurring while users interact with AI systems.

Research Questions:

Can wearable physiological data measure cognitive load during Generative-AI interaction? Can combining physiological and subjective information provide a better understanding of productivity and user well-being?

Key Contribution:

The paper proposes combining wearable physiological measurements with subjective evaluations to study developer experience. This is particularly relevant to FocusBot because sensors could potentially monitor user states and allow an AI desk companion to provide appropriate productivity or break recommendations.

10 **AI and Employee Wellbeing in the Workplace: An Empirical Study**

Link-

Overview:

This 2025 Journal of Business Research study investigates how AI adoption affects employee well-being. The researchers analyzed data from **207 companies** using survey data and structural equation modeling.

Project Objective:

The objective is to understand whether AI directly improves employee well-being or whether its effects occur through improvements in work-related factors. The study specifically examines task optimization and workplace safety.

Research Gap:

The relationship between AI adoption and employee well-being remains relatively under-researched compared with AI's effects on productivity and efficiency. The mechanisms through which AI may influence well-being therefore require further investigation.

Research Questions:

Does AI adoption directly influence employee well-being? Do task optimization, workplace safety and security act as mechanisms through which AI affects employee well-being?

Key Contribution:

The study found that AI adoption did **not directly** affect employee well-being but influenced it

indirectly through task optimization and safety. This supports FocusBot's concept that AI should improve the user's actual working process rather than simply provide more technology.